Laplace Transform.

Wednesday, October 28, 2026 11:32 AM

① Find the Laplace transform of $x(t) = e^{at}u(-t)$.

$$X(s) = \int_{-\infty}^{\infty} e^{at}u(-t)\,e^{-st}\,dt$$

$x(t)$ — e^{-st}: ↓ Kernel corresponding to Laplace transform.



$$= \int_{-\infty}^{0} e^{-(s-a)t}\,dt$$

$$= -\frac{1}{(s-a)}\,e^{-(s-a)t}\Big|_{-\infty}^{0}$$

$$= -\frac{1}{(s-a)}\left[e^{-(s-a)\cdot 0} - e^{-(s-a)\times -\infty}\right]$$

$e^{-(\sigma+j\Omega)t}$ — $|e^{j\Omega}| = 1$

We can only consider a case where $\mathrm{Re}(s) - a < 0$, $\mathrm{Re}(s) < a$

Causes integration to go infinite

$e^{-(s-a)\times -\infty}$ — If this is positive, $s-a > 0$: $= e^{\infty}$

If this is negative, $s-a < 0$ ✓: $= e^{-\infty} = \frac{1}{e^{\infty}} = 0$ ✓

$$= -\frac{1}{(s-a)}\left[1 - \frac{1}{\infty}\right]$$

$$X(s) = -\frac{1}{s-a}[1-0] = -\frac{1}{s-a}, \quad \mathrm{Re}(s) < a$$

$$e^{at}u(-t) \overset{\mathcal{L}}{\longleftrightarrow} -\frac{1}{s-a}, \quad \underbrace{\mathrm{Re}(s) < a}_{\text{ROC}}$$

→ Necessary for specifying when the Laplace integration converges.

② $$x(t) = \begin{cases} e^{-at} & 0 \le t \le T \\ 0, & \text{otherwise} \end{cases}$$

At a quick glance, we see that the signal is a finite-duration signal, so ROC is the entire s-plane.

$$X(s) = \int_0^T e^{-at}e^{-st}\,dt = \int_0^T e^{-(s+a)t}\,dt = -\frac{1}{s+a}e^{-(s+a)t}\Big|_0^T$$

↓ Notice the limits of integral

$$= \frac{-1}{s+a}\left[e^{-(s+a)T} - 1\right] = \frac{1 - e^{-(s+a)T}}{s+a}$$, with ROC as the entire s-plane.

③ $x(t) = e^{2t}u(t) + e^{-3t}u(-t)$ → Compare with the first Q's solution

$a = -3$

ROC 1

e^{-3t} — $\rightarrow -\frac{1}{\quad}$, $\mathrm{Re}(s) < -3$

$a = -3$

$$e^{-3t}u(-t) \longleftrightarrow -\frac{1}{s-(-3)}, \quad Re(s) < -2$$

$$= \frac{-1}{s+3}, \quad Re(s) < -3$$

We got this based on the result from Q1.

Similarly,

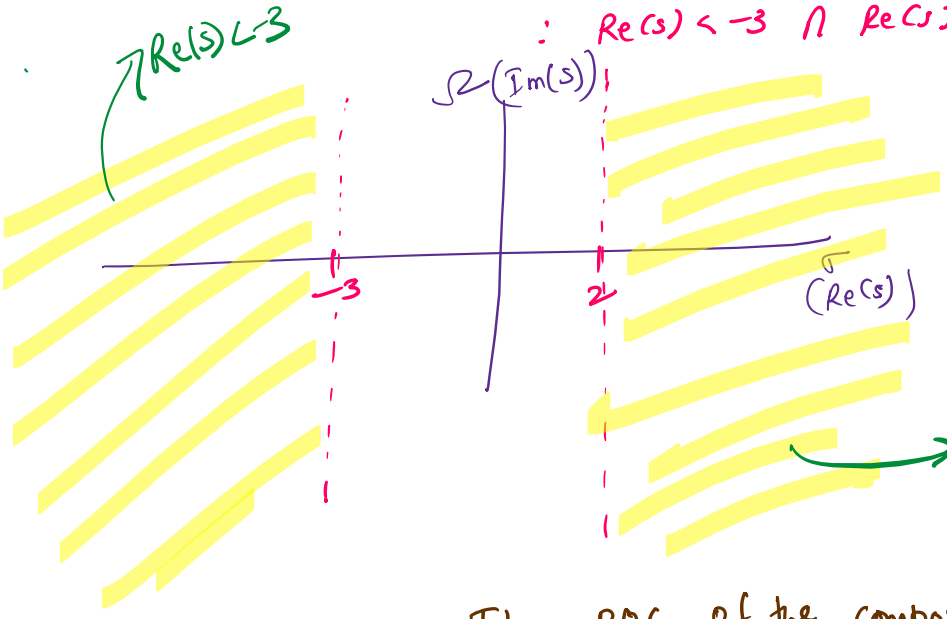
$$e^{at}u(t) \longleftrightarrow \frac{1}{s-a}, \quad Re(s) > a$$ ✓ → we worked on this in slides.

$$e^{2t}u(t) \longleftrightarrow \frac{1}{s-2}, \quad \underbrace{Re(s) > 2}_{RO}$$

Then, we need to check for ROC of the composite signal $e^{2t}u(t) + e^{-3t}u(-t)$

ROC : ROC1 ∩ ROC2

: $Re(s) < -3 \cap Re(s) > 2$



→ These two regions do not overlap

So ROC1 ∩ ROC2 = ∅

If ROC of the composite signal is a null set, we say that no Laplace transform exists.

Thus, $e^{2t}u(t) + e^{-3t}u(-t)$ has no valid Laplace transform.

Q4. $\delta(t) \longleftrightarrow 1$

$x(t) = \delta(t - t_0)$

$X(s) \longleftrightarrow e^{-st_0}$ → for all s.

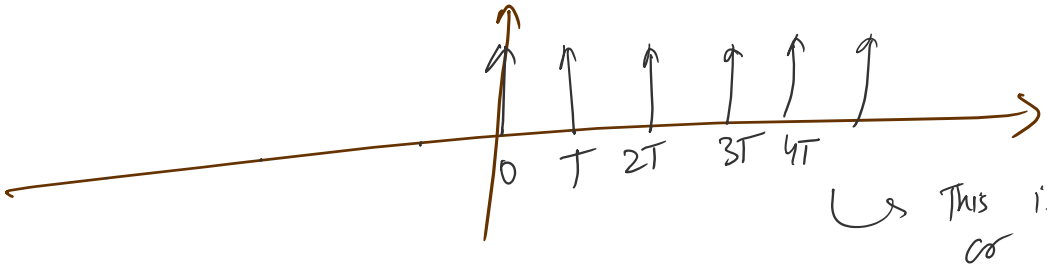
Q5. $x(t) = u(t-t_0)$

$\hat{x}(t) = u(t)$

$\hat{X}(s) \longleftrightarrow \frac{1}{s}$, $Re(s) > 0$

$X(s) \longleftrightarrow \frac{1}{s} e^{-st_0}$, $Re(s) > 0$

Q6. $x(t) = \sum_{k=0}^{\infty} \delta(t-kT)$



↳ This is called impulse train or comb function.

$$X(s) = \sum_{k=0}^{\infty} e^{-skT} = \sum_{k=0}^{\infty} (e^{-sT})^k = \frac{1}{1-e^{-sT}} \quad , Re(s) > 0$$

Q7. Apply time differentiation property to find Laplace transform of $\delta'(t)$. Assume initial condition as zero.

$x(t) \longleftrightarrow X(s)$

$\frac{d}{dt} x(t) \longleftrightarrow sX(s)$

$\delta(t) \longleftrightarrow 1$ for all s

$\delta'(t) = \frac{d}{dt}\delta(t) \longleftrightarrow 1 \cdot s$ for all s.

$\delta'(t) \longleftrightarrow s$ ROC: entire s plane

Q8. $b(t) = e^{-at}\cos(\omega_0 t)\, u(t)$

$\cos(\omega_0 t)\, u(t) = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right) u(t)$

$$d(t) = \cos(\omega_0 t)\,u(t) = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)u(t)$$

$$= \frac{1}{2}e^{j\omega_0 t}u(t) + \frac{1}{2}e^{-j\omega_0 t}u(t)$$

$$D(s) = \frac{1}{2}\left[\frac{1}{s - j\omega_0} + \frac{1}{s + j\omega_0}\right], \quad Re(s) > 0$$

$$= \frac{1}{2}\left[\frac{s + j\omega_0 + s - j\omega_0}{s^2 - (j\omega_0)^2}\right] = \frac{s}{s^2 + \omega_0^2}, \quad Re(s) > 0$$

we use the property $x(t) \longleftrightarrow X(s)$

$$\Rightarrow e^{at}x(t) \longleftrightarrow X(s-a)$$

$b(t) = e^{-at}d(t)$ $\quad$ a is a real number

$$B(s) = D(s+a)$$

$$B(s) = \frac{s+a}{(s+a)^2 + \omega_0^2}, \quad Re(s+a) > 0$$

$$Re(s) > -a.$$

$$dv = e^{-st}dt$$

$$V = \int dv = \int e^{-st}dt$$

$$R(s) = \int_{-\infty}^{\infty} \qquad dt$$

$$\lim_{t \to \infty} -\frac{t}{s}e^{-st}$$ Apply L'hopital Rule

$= \cdot$

$\frac{d}{dt}(te^{-st})$

$(uv)' = u'v + uv'$

$= \lim_{t\to\infty} \frac{d}{dt}(te^{-st})$

$= \lim_{t\to\infty} \left[1 \cdot e^{-st} + t \cdot (-s)e^{-st}\right]$

$= \lim_{t\to\infty} e^{-st}(1 - st)$

$= 0$